



SYSTEM ARCHITECTURE · REFERENCE DOCUMENT

Work Governor

Manoj's AI-powered personal compliance work assistant — driving every application control toward authoritative, system-generated evidence and continuous Argos monitoring.

Next.js 16 (App Router)

React 19

TypeScript

Tailwind CSS 4

Claude Haiku 4.5

Client-side only · No backend DB

Scope

Complete current-state architecture: UI layer, client services, API route, AI tool contract, data model, persistence & file I/O.

Owner

Manoj — personal CORE compliance workspace

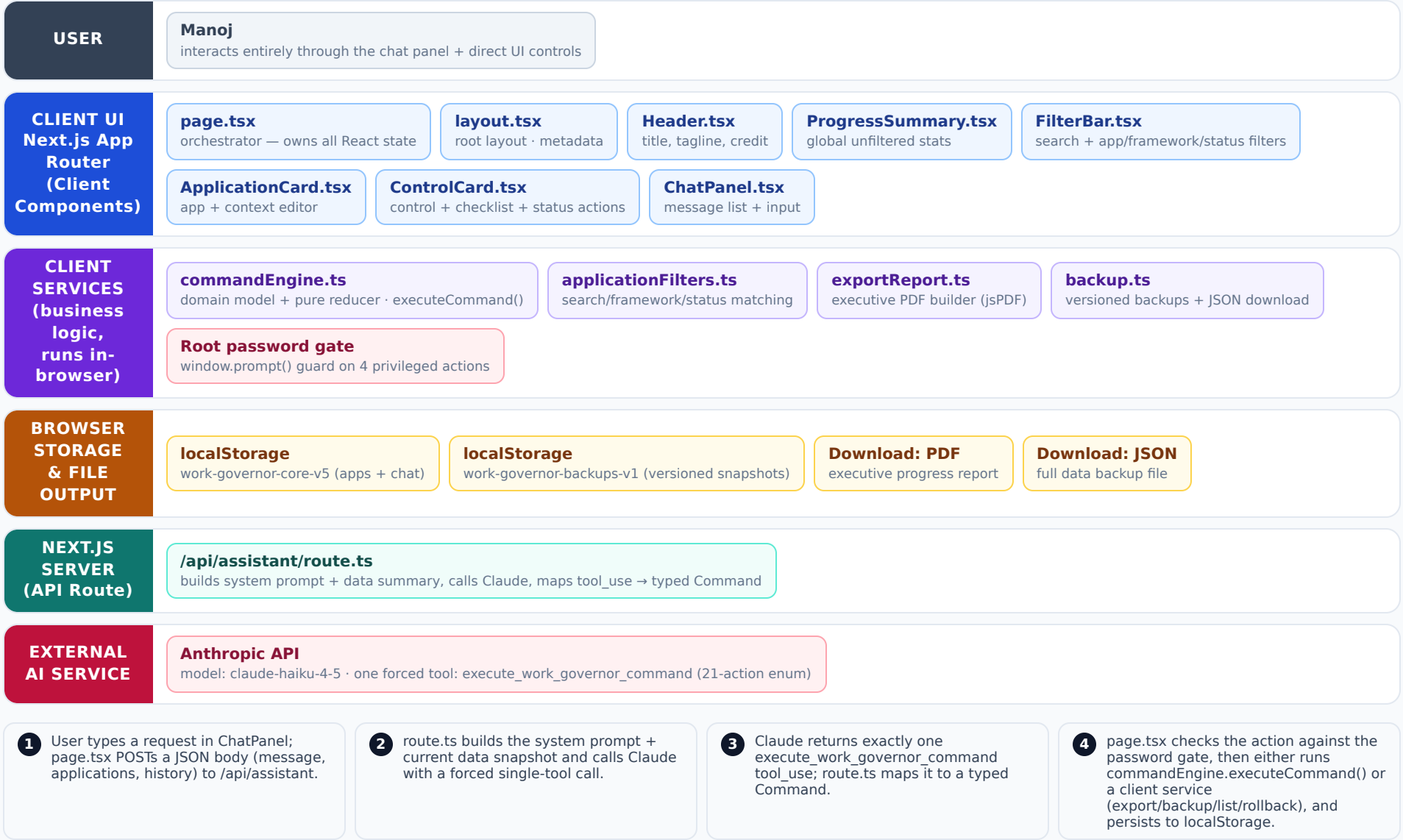
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July 20, 2026

01 · SYSTEM OVERVIEW

High-Level Architecture

Everything runs in the browser except one server route that talks to Claude. No database — the browser *is* the database.



All 21 Chat Actions

Claude always returns exactly one of these 21 actions via a single forced tool call — **PW** marks the 4 password-gated, privileged actions.

AL Application Lifecycle 5 CREATE_APPLICATION Create an app + initial controls; auto-assigns APP-NN only if unnamed UPDATE_APPLICATION_CONTEXT Set purpose, owners, integrations, identity types, etc. RENAME_APPLICATION Change an application's id/name in place DELETE_APPLICATION PW Requires "yes, delete X" confirmation DELETE_ALL_APPLICATIONS PW Requires explicit confirmation	CC Control & Checklist Authoring 3 ADD_CONTROL Add one control with contextual objective/risk/evidence fields REGENERATE_CONTROL_CHECKLIST Rebuild one control's checklist from current context GENERATE_CONTEXTUAL_CHECKLISTS Rebuild every control's checklist for an application	WT Work Tracking 5 UPDATE_HOMework Waiting ↔ Completed on one control UPDATE_CONTROL_WORK General update: status, stage, note, or new tasks UPDATE_ALL_CONTROLS Apply the same update to every control in an app UPDATE_NOTES Append / replace / clear work notes UPDATE_CHECKLIST Append / replace / dedupe & limit checklist tasks
SM Status Management 2 UPDATE_CHECKLIST_STATUS Review Pending / Approved / Needs Revision / Completed UPDATE_CONTROL_STATUS New / In Progress / Ready for Review / Completed / On Hold	DR Data, Reporting & Recovery 4 EXPORT_PROGRESS_REPORT Formatted executive PDF summary (jsPDF) BACKUP_APPLICATION_DATA New numbered snapshot + JSON download LIST_BACKUPS Client composes list from local backup state ROLLBACK_BACKUP PW Confirm, auto-safety-backup, then restore	CM Conversation & Meta 2 CLEAR_CHAT_HISTORY PW Wipes messages only — never app data RESPOND_ONLY Answer / explain / ask for info, no data change

Note: these are not 21 separate Anthropic tools — the API exposes a single tool, `execute_work_governor_command`, whose action field is a 21-value enum. `tool_choice` forces exactly one call per request, so every chat turn produces exactly one command.

Data Model

The entire domain lives in one nested structure — no ORM, no schema migrations, just TypeScript types in commandEngine.ts.

Application1 → many

id, name — identifier

applicationPurpose

applicationOwner

applicationContacts[]

identityTypes[]

dataClassification

contextStatus — derived

hosting — SaaS/PaaS/IaaS/on-prem

businessProcess

technicalOwner

integrations[]

hostingDetails

financialRelevance

controls[] — ComplianceControl

▼ contains many

ComplianceControl1 → many

id, name, framework

controlStatus — derived

controlObjective

applicabilityRationale

argosObjective

nextTasks[] — ChecklistTask

homeworkStatus, stage

checklistStatus

controlRisk

evidenceStrategy

notes[]

▼ contains many

ChecklistTaskleaf node

id, text

completed — boolean

category — TaskCategory

required — boolean

ControlStatus (derived, 6)

New

Checklist Review Pending

In Progress

Ready for Review

Completed

On Hold

ChecklistStatus (4)

Review Pending

Approved

Needs Revision

Completed

WorkflowStage (derived, 6)

Homework

Discovery

Evidence Collection

Testing

Review

Completed

TaskCategory (8)

Homework

Discovery

Access

Evidence Collection

Validation

Approval

Argos Design

Next Steps

ApplicationContextStatus (3)

Missing

Partial

Complete

Framework (2) · HomeworkStatus (2)

SOX

PCI DSS

Waiting

Completed

Technology, Storage & Known Limitations

No backend, no database — a single-user client app with one AI-calling API route.

Technology Stack

● Next.js	16.2.10	App Router, client + server
● React	19.2.4	UI runtime
● TypeScript	^5	end-to-end types
● Tailwind CSS	^4	styling
● @anthropic-ai/sdk	^0.112.3	claude-haiku-4-5 tool calling
● jsPDF + jspdf-autotable	4.2.1 / 5.0.8	PDF generation (client-side)

File I/O — action → output

TRIGGER	OUTPUT FILE
"export progress as a PDF"	work-governor-progress-*.pdf
"take backup now"	work-governor-backup-*.json

Browser Persistence

LOCALSTORAGE KEY	CONTENTS
work-governor-core-v5	applications[] + chat messages[] (auto-migrates from v1-v4)
work-governor-backups-v1	BackupEntry[] — versioned {applications, messages} snapshots

⚠ Known Limitations

- Everything lives in one browser's localStorage — no sync across devices, no server-side database.
- The root password ("123") is a hardcoded client-side placeholder, not real authentication.
- Backups accumulate with no pruning/rotation — fine at personal scale, watch storage limits over time.
- Dead code present in the tree: services/parser.ts, unused components/Header.tsx, layout/Footer.tsx, layout/Sidebar.tsx, plus stray *_BKP files.